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MILITARY SPECIFICATION

RECEIVING SET, SATELLITE SIGNAL, AN/SSR-1A

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for the fabrication, testing, and production of the Receiving Set, Satellite Signal, AN/SSR-1A, hereinafter specified as the receiving set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2.1), the following specifications, standards, and handbooks of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17	Cable, Radio Frequency, Flexible And Semirigid, General Specification For
MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment And Systems, Requirements For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR 003-12), Defense Standardization Program Branch, Department of the Navy, Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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MIL-P-15024	Plates, Tags, And Bands For Identification Of Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication, And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-C-22751	Coating System, Epoxy-Polyamide, Chemical And Solvent Resistant; Process For Application Of
MIL-P-24441	Paint, Epoxy-Polyamide, General Specification For
MIL-M-38510	Microcircuits, General Specification For
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification For
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, Specifications Standards And In Technical Documents
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I-Environmental And Type II-Internally Excited)
MIL-STD-188-100	Common Long Haul And Tactical Communication System Technical Standards
MIL-STD-242	Electronic Equipment Parts Selected Standards, Connectors
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference

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MIL-STD-462 and Interim Notice 5 dated 4 August 1986	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-681	Identification Coding And Application Of Hookup And Lead Wire
MIL-STD-740-1	Airborne Sound Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Testing For Engineering Development, Qualification, And Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
DoD-STD-1399, Section 301	Interface Standard For Shipboard Systems, Section 301, Ship Motion And Attitude
DoD-STD-1399, Section 302	Interface Standard For Shipboard Systems, Section 302, Weather Environment
DoD-STD-1686	Electrostatic Discharge (ESD) Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices)
DoD-STD-2167	Defense System Software Development

HANDBOOKS

MILITARY

MIL-HDBK-175	Microelectronic Device Data Handbook
DoD-HDBK-263	Electrostatic Discharge Control Handbook For Protection Of Electrical And Electronic Parts, Assemblies, And Equipment (Excluding Electrically Initiated Explosive Devices)
MIL-HDBK-781	Reliability Test Methods, Plans, And Environments For Engineering Development, Qualification, And Production

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536E with SCN 1 and SCN 2	Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connections

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NAVAL ELECTRONIC SYSTEMS COMMAND

NAVELEX 0967-LP-541-9010	Receiving Set, Satellite Signal AN/SSR-1 And AN/SSR-1A

NAVAL MATERIAL COMMAND (NAVMAT)

NAVMAT P4855-1	Naval Power Supply Reliability, Design and Manufacturing Guidelines
NAVMATINST 3401.1 (Reference (b))	Naval Material Command Nuclear Survivability Program, Nuclear Hardening Criteria

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-597-1011	Parts Application And Reliability Information Manual For Navy Electronics Equipment

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70-1984	National Electrical Code

(Application for copies should be addressed to National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210).

Copies of specifications, standards, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The requirements specified herein cover a shipboard radio receiving set which, upon installation onboard any type Naval ship, shall provide the capability of receiving teletype broadcasts via satellite relay. The receiving set shall employ multiple (up to four) near-hemispheric antennas so located as to intercept wave fronts from many directions and shall combine the resultant signals in-phase at a single output. The receiving set shall be capable of receiving alternately, but not simultaneously, either frequency modulated (FM) or differentially encoded phase reversal (phase shift keyed (PSK)) signals. FM signals shall be frequency division multiplex (FDM)

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(frequency shift keyed (FSK)) multiplexed teletypewriter (TTY) tones generated by a telegraph terminal modulating the transmitter. PSK signals shall be a serial bit stream containing sixteen 75-bit per second (bps) channels generated by a time division multiplexer (one of the 16 channels is used for frame synchronization). PSK serial data shall be demultiplexed to 16 outputs for TTYs and be compatible with Navy fleet broadcast cryptographic systems. The receiving set shall conform to the requirements of MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.1.2 Receiving set. The receiving set shown in FIGURE 1 is typical and is provided for reference only. Each receiving set shall include four antennas, four amplifier-converters, one combiner-demodulator, and one demultiplexer.

3.1.2.1 Antenna AS-2815/SSR-1 function. The antenna shall receive FM or PSK signals in the Military ultra high frequency (UHF) band from a satellite in a synchronous, equatorial orbit. The receive signal shall be transmitted via coaxial cable to the amplifier-converter.

3.1.2.2 Amplifier-converter AM-6534/SSR-1 function. The amplifier-converter shall receive the UHF signal from the antenna, amplify the signal, and convert it to an intermediate frequency (IF) by mixing it with a local oscillator (LO) frequency received from the combiner-demodulator. The resulting IF signal shall be sent via twin-axial cable to the combiner-demodulator.

3.1.2.3 Combiner-demodulator MD-900/SSR-1 function. The combiner-demodulator shall receive an IF signal from each amplifier-converter, (one to four), amplify each IF signal, and combine the signals in a maximal ratio combiner. The combiner output shall be demodulated in an FM discriminator if the modulation is FM or in a PSK demodulator if PSK modulation is used. The demodulated FM output shall be output to an AN/UCC-1 or similar unit. The demodulated PSK output shall be output to the demultiplexer. The combiner-demodulator shall output the LO frequency to each amplifier-converter. The contractor shall deliver a synthesized local oscillator (SLO) module conforming to the requirements of the APPENDIX with each combiner-demodulator.

3.1.2.4 Demultiplexer TD-1063A/SSR-1 function. The demultiplexer shall separate the 1200 bps serial digital bit stream into 15 75-bps channels and a synchronization word channel. In addition, the demultiplexer shall detect and verify the synchronization word. Synchronization word detection and verification shall provide for repetitive monitoring of the synchronization word in order to make the in-synchronization or out-of-synchronization decision.

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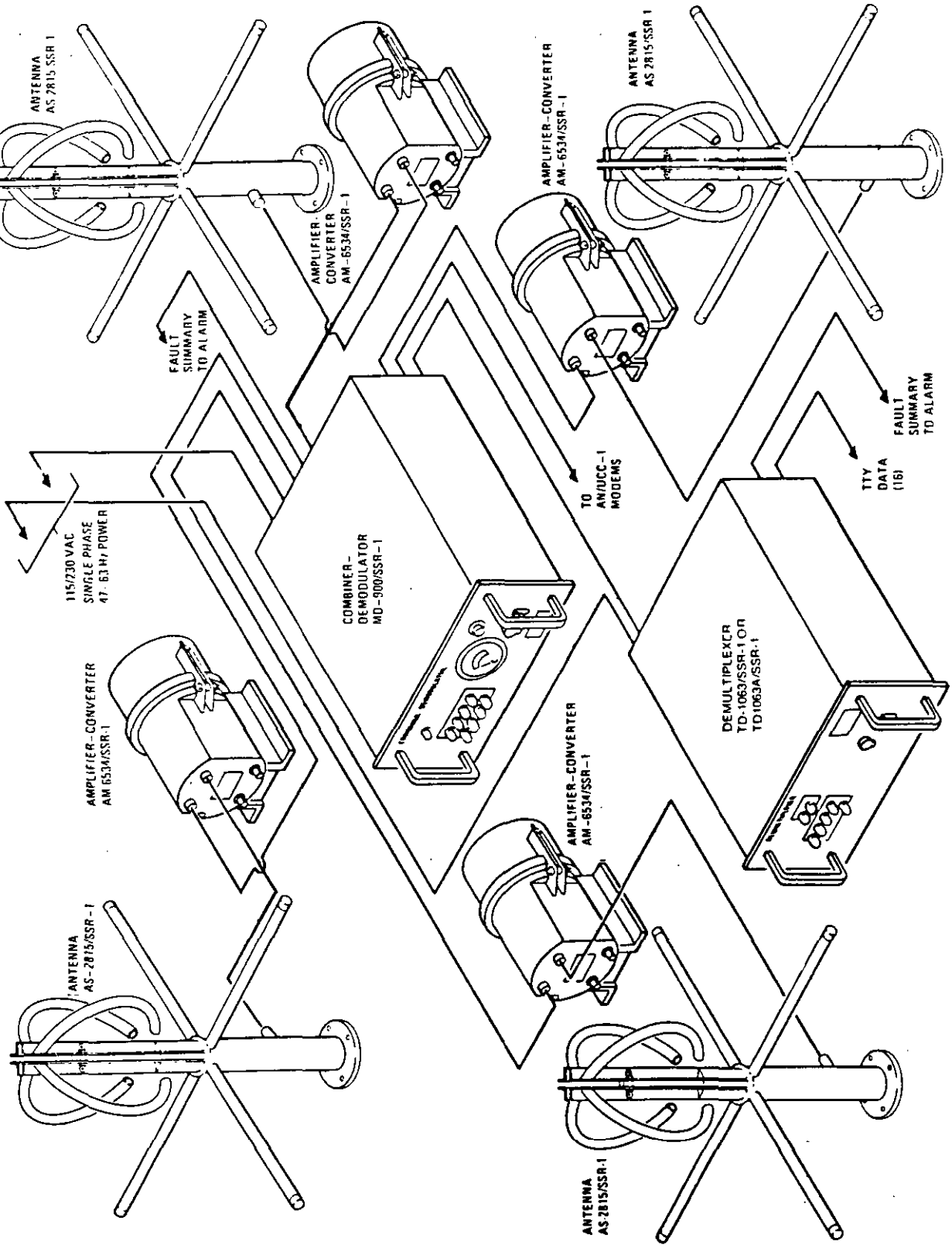


FIGURE 1. Receiving set, satellite signal AN/SSR-1, AN/SSR-1A components.

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3.1.3 Installation requirements. Installation requirements shall be as specified in 3.1.3.1 and 3.1.3.2.

3.1.3.1 Noise environment. The performance of the receiving set shall not be degraded beyond limits specified herein by interference signals generated by collocated continuous wave (CW) transmitters and radars. These interference signals may be in the form of radiated noise power (see 3.2.1.7) or radiated pulse carrier power (see 3.2.4.6).

3.1.3.2 Receiving set location. The individual units of the receiving set shall be located as specified in a through d:

a. Antennas. Exterior hull mounting (up to four units each)
b. Amplifier-converter. Exterior hull mounting (up to three cable-meters (m) (ten cable-feet (ft)) maximum away from antenna) (up to four units each)
c. Combiner-demodulator. Interior (up to 150 cable-m (500 cable-ft) from each amplifier-converter)
d. Demultiplexer. Interior (up to 150 cable-m (500 cable-ft) from the combiner-demodulator)

3.2 Equipment characteristics. Equipment characteristics shall be as specified in 3.2.1 through 3.2.5.12.

3.2.1 Receiving set characteristics. Receiving set characteristics shall be as specified in 3.2.1.1 through 3.2.1.13.

3.2.1.1 Frequency of operation. The operating frequencies shall be from 242.000 megahertz (MHz) to 271.995 MHz in increments of 5 kilohertz (kHz).

3.2.1.2 Receiving set carrier-to-noise (C/N) ratio. A 117-decibels referred to one milliwatt (dBm) CW signal applied at the radio frequency (RF) input of the amplifier-converter shall result in not less than 10 decibels (dB) at 25 degrees Celsius (°C) nor less than 9.5 dB at 50°C C/N ratio as measured at the combiner output. An additional 0.5 dB measurement tolerance is allowed. This signal shall be applied to only one amplifier-converter at a time but all four receiver channels shall be active with three of the four inputs terminated in 50 ohms.

3.2.1.3 C/N enhancement. With the input level to each amplifier-converter adjusted in accordance with 3.2.1.2 to give a C/N ratio of 10 dB ±0.25 dB on each individual channel, a C/N ratio of not less than 15 dB shall be obtained at the combiner output with all four channels operating.

3.2.1.4 Receiving set bandwidth. The receiving set shall have an RF bandwidth of 10 MHz minimum measured at the 1 dB points at the output of the preselector filter, centered at 253.85 MHz.

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3.2.1.5 Bit error rate (BER) with PSK modulation. The maximum BER with PSK modulation shall be 10^{-3} with a signal power level of -127 dBm applied to one receiver channel. This signal shall be applied to one amplifier-converter at a time, but all four receiver channels shall be active with three of the four inputs terminated in 50 ohms.

3.2.1.6 Interconnecting cables. The interconnecting cables between receiving set units shall be as specified in a through c:

a. Between antenna and amplifier-converter. Three m (ten ft) maximum of RG-214 B/U in accordance with MIL-C-17 (not contractor-furnished).

b. Between amplifier-converter and combiner-demodulator. 150 m (500 ft) maximum of RG-22 B/U in accordance with MIL-C-17 (not contractor-furnished). All signals, controls, and power between an amplifier-converter and the combiner-demodulator shall be multiplexed for transmission on a single RG-22 B/U cable in accordance with MIL-C-17.

c. Between combiner-demodulator and demultiplexer. 150 m (500 ft) maximum of RG-108 B/U in accordance with MIL-C-17 (not contractor-furnished)

3.2.1.7 Broadband radiated noise. The receiving set shall be capable of operation without degradation in a broadband radiated noise environment with a received noise power density of -187 dBm per cycle bandwidth.

3.2.1.8 Spurious responses due to intermodulation. With two equal CW signal inputs at levels of -63 dBm or less at frequencies $f_o + \Delta f$ and $f_o \pm 2\Delta f$ where Δf is 1.5 MHz or greater and f_o is the receiver center frequency, the third order spurious response in the receiver shall result in less than 10 dB quieting in the FM mode or 3 dB C/N ratio in the PSK mode.

3.2.1.9 Half-IF response. With a CW signal input at a power level of -76 dBm or less and a frequency of $f_o - \frac{f_{if}}{2}$ where f_o is the receiver center frequency and f_{if} = 19.95 MHz, the half-IF spurious response in the receiver shall result in less than 10 dB quieting in the FM mode or 3 dB C/N ratio in the PSK mode.

3.2.1.10 Image response. With a CW signal input at a power level of -10 dBm or less and frequency of $f_o - 2f_{if}$ where f_o is the receiver center frequency and f_{if} = 19.95 MHz, the image response in the receiver shall result in less than 10 dB quieting in the FM mode or 3 dB C/N ratio in the PSK mode.

3.2.1.11 IF rejection. With a CW input signal at a level of +10 dBm or less at frequency f_{if} = 19.95 MHz the response in the receiver shall result in less than 10 dB quieting in the FM mode or 3 dB C/N ratio in the PSK mode.

3.2.1.12 Desensitization. With a CW input signal at f_o set for 10 dB quieting in the FM mode or 3 dB C/N ratio in the PSK mode, the addition of an

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interfering signal input at a level of -52 dBm or less at a frequency $f_o \pm \Delta f$ where f_o is the receiver center frequency and Δf = 1.5 MHz or greater, the quieting shall remain 9 dB or better in the FM mode or 2.4 dB or better C/N ratio in the PSK mode.

3.2.1.13 Circuit test points. Circuit test point accesses shall be provided in accordance with the Designation, External test points, and Internal test points paragraphs of MIL-STD-415.

3.2.2 Antenna AS-2815/SSR-1 characteristics. Antenna characteristics shall be as specified in 3.2.2.1 through 3.2.2.9.

3.2.2.1 Frequency. The antenna shall receive frequencies between 248.85 MHz and 258.85 MHz.

3.2.2.2 Voltage standing wave ratio (VSWR). The VSWR measured at the RF connector interface shall be less than 2:1 referred to 50 ohms between 248.85 MHz and 258.85 MHz.

3.2.2.3 Polarization. Each antenna shall be right-hand circularly polarized in free space.

3.2.2.4 Antenna pattern. Each antenna shall, in free space, as measured on the appropriate range, provide uniform coverage of 360 degrees (°) in azimuth and 110° in elevation from zenith. The minimum gain of the individual antenna over the frequency range specified shall be as specified in TABLE I. An additional 0.5 dB measurement tolerance will be allowed on specified antenna gain parameters. The minimum gain parameters specified in TABLE I shall be met at the specified elevation and azimuth angles.

TABLE I. Antenna gain.

Elevation angle down from zenith	Minimum gain at four azimuth angles separated by 90° (decibels isotropic (dBi))	Maximum gain degradation from value specified in azimuth column at all other azimuths
0° to 30°	-2.0	No degradation allowed
30° to 70°	-2.0	1.5 dB
70° to 90°	-2.0	3.0 dB
90° to 110°	-5.0	4.0 dB

3.2.2.5 Antenna weight. Maximum weight shall be less than 9 kilograms (kg) (20 pounds (lb)).

3.2.2.6 Antenna mounting. The antenna shall be capable of being bolted by its base to a foundation onboard the ship's hull as specified in the Antenna AS-2815/SRR-1 Installation Drawing figure in NAVELEX 0967-LP-541-9010.

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3.2.2.7 Antenna feedlines. The feedlines from the antenna members to the output connector shall not be exposed.

3.2.2.8 Antenna coaxial cable. A three-m (ten-ft) length of interconnecting coaxial cable with connectors shall be provided with each antenna.

3.2.2.9 Antenna dimensions. Antenna dimensions shall be as specified in the Antenna AS-2815/SSR-1 Installation Drawing figure in NAVELEX 0967-LP-541-9010.

3.2.3 Amplifier-converter AM-6534/SSR-1 characteristics. Amplifier-converter characteristics shall be as specified in 3.2.3.1 through 3.2.3.12.

3.2.3.1 Impedances. The VSWR of the RF input measured at -35 dBm power level shall not exceed 2:1 over the RF bandwidth specified herein. The VSWR of the LO and IF port measured at -5 dBm shall not exceed 2:1 over the LO and IF bandwidths specified herein. The nominal impedance at the RF port shall be 50 ohms. The nominal impedance of the LO and IF port shall be 95 ohms.

3.2.3.2 Frequencies. The frequencies shall be as specified in a through c:

a. RF input frequency. 253.85 MHz ±5 MHz
b. LO input frequency. 222.050 MHz to 252.045 MHz in increments of 5 kHz
c. IF output frequency. 19.95 MHz

3.2.3.3 Signal levels. The signal levels shall be as specified in a through c:

a. RF signal. -117 dBm nominal
b. LO signal. -23 dBm minimum
c. IF output. Conversion gain from RF input to IF output 38 dB ±5 dB measured with RF signal input

3.2.3.4 Preselector filter. An RF preselector filter shall interface with each antenna. The filter shall have the characteristics specified in a through d:

a. Bandwidth at 1.0 dB points shall be not less than 10 MHz
b. Bandwidth at 55 dB points shall be less than 40 MHz
c. Bandwidth at 70 dB points shall be less than 60 MHz
d. The nominal center of the preselector filter bandwidth shall be 253.85 MHz.

3.2.3.5 Isolation. The amplifier-converter shall provide isolation from the antenna of all RF and IF frequencies generated or formed in the receiving set in accordance with the requirements of 3.5.1.9 and 3.5.2.7.

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3.2.3.6 Input termination. An RF switch shall terminate the signal input in 50 ohms for channel gain calibration. The RF switch shall be actuated by a control voltage of 11 volts (V) direct current (DC) (VDC) ±1 VDC supplied from the combiner-demodulator.

3.2.3.7 Input protection. The input shall be protected against burnout by RF levels of 7 V root-mean-square (rms) for thermal protection and 50 V 10 microseconds (μs) RF pulse for peak over-voltage protection. The input levels shall be applied directly to the antenna input port without resultant permanent degradation.

3.2.3.8 Preamplifier. The amplifier-converter shall contain a preamplifier unit which shall provide sideband noise figure as referred to the RF input port of the amplifier-converter of less than 3.0 dB when measured at 25°C, and less than 3.5 dB when measured at 65°C. A ±0.5 dB measurement tolerance will be allowed for testing.

3.2.3.9 Amplifier-converter weight. The maximum weight shall be less than 9 kg (20 lb).

3.2.3.10 Amplifier-converter mounting. The amplifier-converter shall be capable of being bolted to a foundation onboard the ship's hull or internal to the ship's hull as specified in the Amplifier-Converter AM-6534/SSR-1, Installation Drawing figure in NAVELEX 0967-LP-541-9010.

3.2.3.11 Amplifier-converter packaging. The amplifier-converter housing shall be capable of being pressurized with dry nitrogen to resist moisture penetration. Two Schraeder valves compatible with pressurization kit MK-260/U shall be provided. The amplifier-converter shall be capable of containing an internal pressure of 17 psi during all periods of normal operation and storage, and shall be capable of containing an internal pressure of 30 psi during maintenance activities for leak testing. Internal pressure shall not drop below 6 psi from an initial internal pressure of 17 psi over a period of 180 days.

3.2.3.12 Amplifier-converter dimensions. Amplifier-converter dimensions shall be as specified in the Amplifier-Converter AM-6534/SSR-1, Installation Drawing figure in NAVELEX 0967-LP-541-9010.

3.2.4 Combiner-demodulator MD-900/SSR-1 characteristics. Combiner-demodulator characteristics shall be as specified in 3.2.4.1 through 3.2.4.17.

3.2.4.1 Impedances. The four IF and LO interface ports shall not exceed a VSWR of 2:1 referred to 95 ohms measured at a power level of -5 dBm at the IF and LO frequencies.

3.2.4.2 Isolation. Isolation between the interface ports to the amplifier-converter at the IF shall be 30 dB minimum.

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3.2.4.3 Interface ports signal characteristics. The signals passing between the interface ports and the amplifier-converter shall have the characteristics specified in 3.2.4.3.1 through 3.2.4.3.4.

3.2.4.3.1 LO output from combiner-demodulator. LO output from the combiner-demodulator shall be as specified in a and b:

a. Balanced between conductors of twin-axial cable
b. Output level of +17 dBm minimum (see 3.2.4.4)

3.2.4.3.2 IF input to combiner-demodulator. IF input to the combiner-demodulator shall be as specified in a and b:

a. Balanced between conductors of twin-axial cable
b. Nominal frequency of 19.95 MHz

3.2.4.3.3 Amplifier-converter DC power from combiner-demodulator. Amplifier-converter DC power from the combiner-demodulator shall be as specified in a through c:

a. Carried on one conductor of twin-axial cable. Return through shield.
b. 20 VDC $\pm$4 VDC
c. Short-circuit protected

3.2.4.3.4 Amplifier-converter input termination control voltage. The amplifier-converter input termination control voltage shall be as specified in a through c:

a. Carried on second conductor of twin-axial cable. Return through shield.
b. Short-circuit protected
c. 11 VDC $\pm$1 VDC

3.2.4.4 LO characteristics. LO characteristics shall be as specified in the APPENDIX.

3.2.4.5 Channel units. The combiner-demodulator shall contain four identical channels which shall contain filters, blanking units, and any other circuitry required to condition, amplify, adjust, or otherwise manipulate signals from each amplifier-converter unit prior to combining. Individual channel gain adjustment range shall be 10 dB minimum.

3.2.4.6 Pulse blanking. Prior to combining, in-band pulse interference shall be removed by blanking the appropriate channel during the period of the pulse. This blanking shall take place without receipt of information from sources external of the receiving set. Pulses with pulsewidths from 10 μs to 500 μs and duty cycle of less than 20 percent shall be blanked with a delay time less than 5 μs. Blanking action shall occur when the pulse-to-signal amplitude ratio exceeds 20 dB $\pm$10 dB.

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3.2.4.6.1 Pulse blanking bypass. A switch shall be provided within the combiner-demodulator enclosure to enable the operator to disable the pulse blanking function.

3.2.4.7 In-band overload protection. Prior to combining, in-band average power shall be continuously measured and that channel inhibited when an input level to the amplifier-converter of -90 dBm +10 dB, -0 dB is exceeded.

3.2.4.8 Signal combiner. The combiner-demodulator shall contain a signal combiner which shall coherently sum the desired signals from all four channel units using maximal ratio techniques. The combiner shall operate with modulation formats that totally remove the carrier (phase reversal keying), that have components near the carrier (digital PSK or multiple frequency shift keying (MFSK)), or that have full carrier.

3.2.4.9 Demodulation unit. The demodulation unit shall contain two demodulator circuits (FM and PSK) only one of which will be operative at a given time. Change of demodulation operation modes shall be accomplished by switch activation.

3.2.4.9.1 FM demodulator. The demodulator shall accept the IF signal from the combiner, detect the FM signal, and supply the proper baseband output.

3.2.4.9.2 FM characteristics. Up to 16 FSK TTY tone keyers in a 300 Hertz (Hz) to 3400 Hz voice channel shall be used to modulate the carrier. The deviation will be such that the significant sidebands are contained within $\pm$10 kHz of the carrier.

3.2.4.9.3 FM output level. The FM baseband output level shall be continuously adjustable over at least a -25 dBm to +10 dBm range by a module-mounted adjustment. Short-circuit protection shall be provided.

3.2.4.9.4 FM output impedance. The FM output shall be transformer coupled and balanced. It shall have a nominal impedance of 600 ohms independent of output level, longitudinal balance of 40 dB minimum and return loss of 20 dB minimum.

3.2.4.9.5 FM output bandwidth. The baseband output shall conform to the Parameters for Nominal 4-kHz Voice Bandwidth Channel paragraph of MIL-STD-188-100.

3.2.4.9.6 FM distortion. The distortion shall conform to the Parameters for Nominal 4-kHz Voice Bandwidth Channel paragraph of MIL-STD-188-100

3.2.4.9.7 FM pre-emphasis and de-emphasis. The demodulator shall have no de-emphasis network that complements any pre-emphasis network employed in the transmitter.

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3.2.4.9.8 PSK demodulator. The PSK demodulator shall accept the IF output of the combiner and coherently detect the modulation and supply the serial data stream to the demultiplexer.

3.2.4.9.9 PSK synchronization. The PSK demodulator shall accomplish bit synchronization automatically without need for the transmission of separate synchronization signals from the modulator.

3.2.4.9.10 PSK output level. The output level shall be 1 V peak-to-peak minimum.

3.2.4.9.11 PSK output impedance. The balanced output shall have a nominal impedance of 600 ohms, a longitudinal balance of 40 dB minimum, and a return loss of 20 dB minimum.

3.2.4.10 Amplifier-converter DC power control. Means shall be provided in the combiner-demodulator to remove DC power from each of the individual amplifier-converter units. If DC power is removed from an amplifier-converter, an indicator on the front panel of the combiner-demodulator shall be illuminated indicating a channel is disabled.

3.2.4.11 Signal monitor. A relative level indicating meter shall be provided to monitor the parameters specified in a through d:

a. Automatic gain control (AGC) voltage
b. LO frequency relative to carrier frequency
c. Signal levels of all four channels
d. FM output level

The signal monitored shall be selectable by switch. A phone jack shall be provided to monitor the data output or the FM tone package.

3.2.4.12 Fault indicators. Fault indicators shall be provided on the levels specified in a and b:

a. Level I. A simple drawer status indication shall be provided by relay contacts (open-go; closed-no-go)

b. Level II. Visual fault indicators shall be provided on the front panel. These indicators shall, where practical, permit the technician to isolate most malfunctions to the module or line replaceable unit (LRU) without disassembly or trial and error replacement.

Visual fault indicators for the amplifier-converter shall be included on the combiner-demodulator front panel.

3.2.4.13 Elapsed time meter. An elapsed time meter shall be provided in the combiner-demodulator in accordance with Requirement 51 of MIL-STD-454. The meter shall be connected to the alternating current (AC) power.

3.2.4.14 Combiner-demodulator weight. The maximum weight of the combiner-demodulator shall be less than 40.5 kg (90 lb).

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3.2.4.15 Combiner-demodulator container. The combiner-demodulator shall include outer and inner containers as specified in a through f:

a. The outer container shall be suitable for standard 48.3-centimeter (cm) (19-inch) rack mounting or for bench mounting at the option of the installer.

b. The inner container shall be extendable from the outer container by means of chassis slides of the withdrawal, tilt, and lock type. Lock positions shall permit easy access to circuit elements in both top and bottom of the container.

c. Both inner and outer containers shall provide protection for fixtures and connectors such that a container may be placed on any of its faces.

d. There shall be a cable retract mechanism between inner and outer container connectors. All connections to exterior units shall be made at the rear of the outer container.

e. The outer container shall be totally enclosed except for front opening.

f. The inner container shall have two handles such that the entire weight of both containers may be supported by either handle alone and that the inner container may be locked to the outer container with captive slotted thumb screws. The width of the handles shall be at least 1.27 cm (0.50 inch) with no sharp edges.

3.2.4.16 Drawer harnesses and cables. Drawer harnesses and cables shall be capable of being replaced without unsoldering and soldering.

3.2.4.17 Combiner-demodulator dimensions. Combiner-demodulator dimensions shall be as specified in the Combiner-Demodulator MD-900/SSR-1, Installation Drawing figure in NAVELEX 0967-LP-541-9010.

3.2.5 Demultiplexer TD-1063A/SSR-1 characteristics. Demultiplexer characteristics shall be as specified in 3.2.5.1 through 3.2.5.12.

3.2.5.1 Input impedance. The input impedance shall be a balanced 10,000 ohms ±20 percent.

3.2.5.2 Input level. The demultiplexer shall be capable of operating with input levels as specified in the Low Level Digital Interfaces paragraph of MIL-STD-188-100.

3.2.5.3 Bit integrity. Frame synchronization and bit integrity shall not be lost when signal fades (below demodulator threshold) of up to 10 seconds occur.

3.2.5.4 Output circuitry. The demultiplexer is capable of providing either high-level or low-level interface outputs, dependent on which set (2 each) of TTY switch driver modules is installed. Characteristics of the high-level interface outputs are specified in 3.2.5.4.1 through 3.2.5.4.4. Characteristics of the low-level interface outputs are specified in 3.2.5.4.5. Unless otherwise specified (see 6.2.1), low-level TTY switch drivers shall be supplied.

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3.2.5.4.1 Keying. Each of the 15 channels shall be capable of keying a 20 to 120 milliampere (mA) neutral telegraph loop (120 VDC to 150 VDC battery externally supplied) at 75 bps with neither bias nor distortion exceeding five percent.

3.2.5.4.2 Additional capacitance. The receiving set shall continue to operate as specified in 3.2.5.4.1 with the addition of 2 microfarad (μF) capacitance across the two-wire output with no degradation except that bias or distortion shall not exceed 25 percent.

3.2.5.4.3 Mark and space output. For a mark output, the keying circuit shall have a series resistance no greater than 150 ohms. For a space output, the keying circuit shall have a series resistance of not less than 100,000 ohms.

3.2.5.4.4 Protection against circuit failure. Operation shall not be degraded by inductive transients of 20 millijoules (mJ) of energy and 300 V peaks. Protection against circuit failure (other than blown fuses) shall be provided in cases where loop currents in excess of 120 mA and loop transient voltages (normal or reverse) of up to 300 V occur.

3.2.5.4.5 Low-level outputs. When low-level TTY switch driver modules are installed, each of the 15 channel outputs shall be +6.0 V $\pm$1.0 V peak for a space output and -6.0 V $\pm$1.0 V peak for a mark output.

3.2.5.5 Signal monitor. A phone jack shall be provided to monitor the data input.

3.2.5.6 Fault indicators. Fault indicators shall be provided on the levels specified in a and b:

a. Level I. A simple drawer status indication shall be provided by relay contacts (open-go; closed-no-go).

b. Level II. Visual fault indicators shall be provided on the front panel. These indicators shall, where practical, permit the technician to isolate most malfunctions to the module or LRU without disassembly or trial and error replacement.

3.2.5.7 Status indicators. Visual status indicators shall be provided on the front panel. These indicators shall illuminate if there is no data input and also if the demultiplexer is searching for frame synchronization.

3.2.5.8 Elapsed time meter. An elapsed time meter shall be provided in the demultiplexer in accordance with Requirement 51 of MIL-STD-454. The meter shall be connected to the AC power.

3.2.5.9 Demultiplexer weight. The maximum weight of the demultiplexer shall be less than 36 kg (80 lb).

3.2.5.10 Demultiplexer container. The demultiplexer shall include outer and inner containers as specified in a through f:

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a. The outer container shall be suitable for standard 48.3-cm (19-inch) rack mounting or for bench mounting at the option of the installer.

b. The inner container shall be extendable from the outer container by means of chassis slides of the withdrawal, tilt, and lock type. Lock positions shall permit easy access to circuit elements in both top and bottom of the container.

c. Both inner and outer containers shall provide protection for fixtures and connectors such that a container may be placed on any of its faces.

d. There shall be a cable retract mechanism between inner and outer container connectors. All connections to exterior units shall be made at the rear of the outer container.

e. The outer container shall be totally enclosed except for front opening.

f. The inner container shall have two handles such that the entire weight of both containers may be supported by either handle alone and that the inner container may be locked to the outer container with captive slotted thumb screws. The width of the handles shall be at least 1.27 cm (0.50 inch) with no sharp edges.

3.2.5.11 Drawer harnesses and cables. Drawer harnesses and cables shall be capable of being replaced without unsoldering and soldering.

3.2.5.12 Demultiplexer dimensions. Demultiplexer dimensions shall be as specified in the Demultiplexer TD-1063/SSR-1, TD-1063A/SSR-1, Installation Drawing figure in NAVELEX 0967-LP-541-9010.

3.2.6 Receiving set RF and electrical interfaces. Receiving set RF and electrical interfaces shall be as specified in the Receiving Set, Satellite Signal AN/SSR-1, AN/SSR-1A, Overall Block Diagram figure and the Receiving Set, Satellite Signal AN/SSR-1, AN/SSR-1A, Interconnection Diagram figure of NAVELEX 0967-LP-541-9010, and as specified in 3.2.6.1 through 3.2.6.4.

3.2.6.1 Antenna interface. The antenna interface shall be as specified in the Antenna AS-2815/SSR-1 Installation Drawing figure and the Amplifier Converter and Combiner-Demodulator Functional Signal-Flow Diagram figure of NAVELEX 0967-LP-541-9010.

3.2.6.2 Amplifier-converter interfaces. The amplifier-converter interfaces shall be as specified in the Amplifier-Converter AM-6534/SSR-1, Installation Drawing figure and the Amplifier Converter and Combiner-Demodulator Functional Signal-Flow Diagram figure of NAVELEX 0967-LP-541-9010.

3.2.6.3 Combiner-demodulator interfaces. The combiner-demodulator interfaces shall be as specified in the Combiner-Demodulator MD-900/SSR-1, Installation Drawing figure and the Amplifier Converter and Combiner-Demodulator Functional Signal-Flow Diagram figure of NAVELEX 0967-LP-541-9010.

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3.2.6.4 Demultiplexer interfaces. The demultiplexer interfaces shall be as specified in the Demultiplexer TD-1063/SSR-1, TD-1063A/SSR-1, Installation Drawing figure and the Demultiplexer Functional Signal-Flow Diagram figure of NAVELEX 0967-LP-541-9010.

3.3 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (Θ_0 as defined by MIL-HDBK-781) shall be 3000 hours. The lower test MTBF (Θ_1 as defined by MIL-HDBK-781) shall be 1500 hours.

3.4 Maintainability. The receiving set shall have a specified equipment repair time (ERT) no greater than 15 minutes when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit (PC) cards) and chassis-mounted components (including electronic, electrical, electromechanical, and mechanical parts). Corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance.

3.5 Environmental requirements. The above-deck and below-deck equipment shall conform to the requirements specified in 3.5.1 through 3.5.2.12.

3.5.1 Above-deck equipment. The above-deck equipment (antennas and amplifier-converters) shall conform to the environmental requirements specified in 3.5.1.1 through 3.5.1.14.3.

3.5.1.1 Temperature range. The above-deck equipment shall operate from -28°C to +65°C and have a nonoperating (storage) range of -62°C to +71°C.

3.5.1.2 Salt fog. The above-deck equipment shall be resistant to the corrosive effects of salt sea atmosphere.

3.5.1.3 Altitude. The above-deck equipment shall survive nonoperating storage and transport in a low pressure environment equivalent to 4500 m (15000 ft) above sea level. The above-deck equipment shall operate normally at pressure equivalent to 1500 m (5000 ft) above sea level.

3.5.1.4 Wind. The above-deck equipment shall operate normally in wind speeds up to 100 knots.

3.5.1.5 Ice loading. The above-deck equipment shall operate normally with ice loading of up to 2.0 kg (4.5 lb) per square foot.

3.5.1.6 Sea state. The above-deck equipment shall operate normally under sea state five as specified in DoD-STD-1399 Section 301 and Section 302. The above-deck equipment, nonoperating, shall survive sea state eight.

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3.5.1.7 Shock. The above-deck equipment shall conform to the shock requirements of MIL-S-901 for Grade A, Class 1, Type A.

3.5.1.8 Vibration. The above-deck equipment shall conform to the vibration requirement of MIL-STD-167-1, Type I. The maximum frequency shall be 50 Hz. The equipment shall comply with the vibration requirements without internal or external resilient mountings.

3.5.1.9 Electromagnetic compatibility. The above-deck equipment shall be self-compatible and meet the following requirements of MIL-STD-461 for Class A4 equipment: CE01, CE03, RE01, RE02, RS01, RS02, RS03, CS01, CS02, CS03, CS04, CS05, and CS06 (at 250 V spike).

3.5.1.10 Enclosure. The above-deck equipment shall be spraytight in accordance with MIL-STD-108.

3.5.1.11 Humidity. The equipment shall withstand humidity environments up to 95 percent maximum.

3.5.1.12 Inclination. The above-deck equipment shall conform to the Inclination paragraph of MIL-E-16400 for a 45° inclination angle.

3.5.1.13 Electromagnetic pulse (EMP) hardening. The above-deck equipment shall be hardened or shall be intrinsically hard against the EMP resulting from on exoatmospheric nuclear burst. The equipment shall not exhibit any permanent malfunction or degradation of any requirements of the equipment as specified herein, after being subjected to the RS05 test signal requirement of MIL-STD-461 for Class A4 equipment.

3.5.1.14 Nuclear effects. The above-deck equipment shall conform to the requirements of 3.5.2.14.1 through 3.5.1.14.3.

3.5.1.14.1 Transient radiation effects on electronics (TREE). The above-deck equipment shall operate under a TREE environment and survive the nuclear hardness levels specified in Enclosure (1) to Ref (b) of NAVMATINST 3401.1 (18 April 1985) without permanent damage, or interruption of service beyond the duration of the transient. Degradation due to neutrons or total dose shall not cause the equipment to perform below allowable limits for operation as specified herein.

3.5.1.14.2 Nuclear air blast. The above-deck equipment shall comply with the nuclear air blast requirements as specified in Enclosure (1) to Reference (b) of NAVMATINST 3401.1 (18 April 1985) without permanent damage, or interruption of service beyond the duration of the air blast.

3.5.1.14.3 Nuclear thermal radiation exposure. The above-deck equipment shall comply with the nuclear thermal radiation requirements as specified in

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Enclosure (1) to Reference (b) of NAVMATINST 3401.1 (18 April 1985) without permanent damage, or interruption of service beyond the duration of the thermal radiation.

3.5.2 Below-deck equipment. The below-deck equipment (combiner-demodulator and demultiplexer) shall conform to the environmental requirements specified in 3.5.2.1 through 3.5.2.12.

3.5.2.1 Temperature range. The below-deck equipment shall operate from 0°C to +50°C and have a nonoperating (storage) range of -62°C to +71°C.

3.5.2.2 Salt fog. The below-deck equipment shall be resistant to the corrosive effects of salt sea atmosphere.

3.5.2.3 Altitude. The below-deck equipment shall survive nonoperating storage and transport in a low pressure environment equivalent to 4500 m (15000 ft) above sea level. The below-deck equipments shall operate normally at pressure equivalent to 1500 m (5000 ft) above sea level.

3.5.2.4 Sea state. The below-deck equipment shall operate normally under sea state five as specified in DOD-STD-1399, Section 301 and Section 302. The below-deck equipment shall survive, nonoperating, sea state eight.

3.5.2.5 Shock. The below-deck equipment shall conform to the shock requirements of MIL-S-901 for Grade A, Class 1, Type A.

3.5.2.6 Vibration. The below-deck equipment shall conform to the vibration requirement of MIL-STD-167-1, Type I. The maximum frequency shall be 50 Hz. The equipment shall comply with the vibration requirements without internal or external resilient mountings.

3.5.2.7 Electromagnetic compatibility. The below-deck equipment shall be self-compatible and meet the following requirements of MIL-STD-461 for Class A4 equipment: CE01, CE03, CE06, RE01, RE02, RS01, RS02, RS03, CS01, CS02, CS03, CS04, CS05, CS06 (at 250 V spike).

3.5.2.8 Enclosure. The below-deck equipment shall be drip proof to 45° in accordance with MIL-STD-108.

3.5.2.9 Humidity. The equipment shall withstand humidity environments up to 95 percent maximum.

3.5.2.10 Inclination. The below-deck equipment shall conform to the Inclination paragraph of MIL-E-16400 for a 45° inclination angle.

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3.5.2.11 EMP hardening. The below-deck equipment shall be hardened or shall be intrinsically hard against the EMP resulting from on exoatmospheric nuclear burst. The equipment shall not exhibit any permanent malfunction, degradation of performance, or deviation from any requirement of the equipment specified herein after being subjected to the CS11 test signal requirement of MIL-STD-461 for Class A4 equipment.

3.5.2.12 Nuclear TREE hardening. The below-deck equipment shall operate under a TREE environment and survive the nuclear hardness levels specified in Enclosure (1) to Reference (b) of NAVMATINST 3401.1 without permanent damage or interruption of service beyond the duration of the transient. Degradation due to neutrons or total dose shall not cause the equipment to perform below allowable limits for operation specified herein.

3.6 Parts, materials, and processes. The receiving set shall conform to the requirements specified in 3.6.1 through 3.6.8.

3.6.1 Selection of parts. Selection of parts, materials, processes and standards shall be in accordance with MIL-E-16400 and as specified herein.

3.6.2 Parts control. The parts to be incorporated in the receiving set shall be controlled in accordance with MIL-STD-965, Procedure I.

3.6.3 Additional equipment. As a minimum, the contractor shall furnish the additional equipment specified in TABLE II. The equipments specified in TABLE II are required for shipboard scheduled maintenance on the receiving set.

TABLE II. Additional equipment.

Equipment	Identification
Attenuator 1/	99899-4772-30
SMA-to-BNC adapter	FSN 5935-401-7228
Tuning tool	73899-7104-5

1/ Frequency: 228 MHz to 238 MHz; Power: 1 watt (W) minimum; Attenuation: 30 dB

3.6.4 Amplifier-converter. The contractor shall ensure and provide proof to the Government that the cover holding bolts for the amplifier-converter are corrosion-resistant steel (CRES). Two spare O-rings and two spare RF gaskets shall be included inside each sealed amplifier-converter unit.

3.6.5 Connectors and adapters. A complete set of mating halves of connectors required for installation shall be provided. Positioning keys in Type RG-22 connectors of MIL-C-39012 shall be welded (not press fit). Means shall be taken to preclude damage due to cable weight or partially inserted connectors.

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3.6.6 Parts rescreening. All semiconductor, microelectronic, and hybrid devices shall be rescreened at -55°C, +25°C, and +125°C. The applicable device specification parameters for each device shall be measured at each temperature.

3.6.7 Metal parts. Metal parts used in fabrication of this equipment shall be as specified in the Metals and alloys paragraph of MIL-E-16400.

3.6.8 Fungus. The equipment shall be resistant to fungi and shall not be adversely affected by fungi under conditions favorable for fungus development (for example, high humidity, warm atmosphere and presence of inorganic salt).

3.7 Design and construction. The receiving set shall conform to the requirements specified in 3.7.1 through 3.7.25.

3.7.1 Accessibility. Exterior doors, hinged covers, and slide drawers shall have automatic latch and quick release devices which must be actuated before they can be opened, and stops to lock them in the open position for each maintenance configuration. The design shall be such that sufficient clearance is provided in each maintenance configuration to allow required tasks to be performed without undue discomfort, danger, or effort. Access doors, if required, shall be nonstructural and installed with the minimum number of captivated fasteners. Easy access to all components for maintenance shall be provided to the maximum extent practicable. The controls shall be readily accessible and simple to operate. The receiving set shall be maintainable by a two-person crew over any 12 hour period.

3.7.2 Maintenance. The maintenance concept for repair of the receiving set shall be organizational replacement of plug-in type modules and printed wiring boards (PWB). Components and piece parts incident to, but not part of, modules or PC boards may be replaced by a technician. Circuits shall be modularized so that similar functions shall use interchangeable modules. Connectors shall be located and spaced to allow ease of handling by maintenance personnel wearing gloves.

3.7.2.1 Interchangeability. LRUs shall be completely interchangeable. After replacing an LRU, the only adjustments allowed to be made are those specified in NAVELEX 0967-LP-541-9010, Chapter 6, Section I. The equipment shall be manufactured such that forward compatibility and backward compatibility is achieved between LRUs in delivered units and previously-delivered fleet units.

3.7.3 Safety. All construction and material installation performed by the contractor shall be designed to prevent injury to personnel or equipment during installation, operation, and maintenance. The personnel safety shall conform to the Safety (personnel hazard) paragraph of MIL-E-16400.

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3.7.3.1 Critical controls. Critical controls, the accidental activation of which may cause damage to equipment, injury to personnel, or degradation of system function, shall be designed and located so that they are not susceptible to being moved accidentally.

3.7.3.2 Noise, hearing conservation. Sound levels produced by the receiving set shall not exceed an A-weighted sound pressure level of 80 dB when measured 0.9 m (3 ft) from the equipment.

3.7.3.3 Weight marking. Equipment weighing in excess of 15.8 kg (35 lb) shall have the weight clearly marked on the external surface of the equipment and readily visible during installation and removal. The marking shall be as permanent as the normal life expectancy of the unit to which it is applied.

3.7.3.4 Leakage current. The leakage current of the receiving set as measured shall not exceed 5 mA at 115 V or 230 V, 60 Hz, single-phase source power.

3.7.3.5 115 VAC equipment power connections. 115 VAC power cable connections within the equipment shall have the conductor and color assignments specified in TABLE III. When connectors are used, the pin designation shall be as specified in TABLE III. Black paired with white or brown paired with light blue conductor insulation color shall be maintained from the equipment input connections to all components having the same voltage and frequency as the input power.

TABLE III. 115 VAC equipment power connections.

Pin designation	Conductor assignment	Insulation color code	
		National	International
A	115 VAC return	White	Light blue
B	Safety ground	Green	Green with yellow stripe
C	115 VAC high (supply)	Black	Brown

3.7.3.6 Drawer slides. Drawer slide design shall include provisions to prevent accidental derailing and detachment of equipment from the slider.

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3.7.3.6.1 Support strength. Drawer slides in the closed and extended positions, detentes, mounting surfaces, and other attachment mechanisms which support equipment shall have a safety factor equal to twice the maximum anticipated weight.

3.7.4 PWB fabrication and PWB protection. PWB fabrication and protection shall be as specified in 3.7.4.1 and 3.7.4.2.

3.7.4.1 PWBs. PWBs shall be fabricated in accordance with the requirements of MIL-STD-275 and MIL-P-55110. PWBs and chassis shall be marked to key and facilitate insertion of the PWB into the correct position. No damage shall occur if a PWB is inserted into an incorrect position.

3.7.4.2 PWB protection. With power applied to the equipment, removal or insertion of each PWB shall not damage the PWB or any other part of the equipment. Each PWB shall be keyed to prevent incorrect insertion.

3.7.5 Power supply protection. Fault conditions ranging from open-circuits to short-circuits shall cause no damage to the power supply.

3.7.6 Electrostatic discharge protection requirements. Components and modules with inherent susceptibility to the effects of electrostatic discharge shall be properly protected in conformance to the requirements of DoD-STD-1686 and DoD-HDBK-263.

3.7.7 Material flammability. Material used as part of the equipment shall conform to Requirement 3 of MIL-STD-454.

3.7.8 Overcurrent protection. Electrical overcurrent protection shall conform to the Circuit breakers, Fuses and fuseholders, and Electrical overload protection paragraphs of MIL-E-16400. When fuses are used, front panel blown fuse indicators shall be provided. When circuit breakers are used, a visible status indication shall be provided.

3.7.8.1 Overcurrent device location. A fuse shall be connected in series with each ungrounded power conductor as determined by the NFPA 70-1984 for grounded source power and shall be located on the load side of the power switch.

3.7.9 Thermal design. Equipment thermal design shall be in accordance with the paragraphs of MIL-E-16400 specified in a through e:

a. Thermal design
b. Heat removal
c. Forced air-cooling
d. Ventilation openings
e. Cooling during maintenance

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If equipment forced air cooling is required, all air intakes shall be covered by a suitable air filter which can be cleaned with soap and water.

3.7.9.1 High temperature protection. The equipment shall employ fail-safe devices which protect the equipment from damage in the event of loss of cooling air or in the event of excessively high ambient temperatures. The fail-safe devices shall provide protection while the equipment is in the standby mode. An override capability shall be provided.

3.7.10 Equipment electrical power on-off switch. A switch or circuit breaker for disconnecting equipment from all electrical power systems shall be mounted on the equipment front panel and its function clearly labeled. It shall break all power conductors of the power circuit.

3.7.11 Handles. Handles shall be provided on the front panel of the equipment to ensure safe and easy removal of the equipment from the mounting. The handles shall have no sharp edges. Handles shall not protrude more than 5.08 cm (2 inch) from the front panel surface. All controls on the front panel shall not protrude beyond the handles.

3.7.12 Locking devices. Locking devices or materials required on internal adjustment controls to prevent the need for readjustment of the controls after sustained periods of shock and vibration shall conform to MIL-STD-454, Requirement 12 and Requirement 28.

3.7.13 Mounting holes. All tapped aluminum mounting holes (except grounding points and pipe threads) shall contain inserts.

3.7.14 Finish. The exterior of the antenna and amplifier-converter units shall be painted gray in accordance with MIL-P-24441. The exterior of the combiner-demodulator and demultiplexer units shall be painted in accordance with MIL-E-15090, Class 2 (semi-gloss), Type III (baking) enamel. Painting of the front panel and edges of the equipment shall be in accordance with the Painting paragraph of MIL-E-16400. Epoxy paint conforming to MIL-C-22751 may be used in lieu of enamel. The remaining surfaces of the equipment shall be treated with chemical film in accordance with MIL-C-5541.

3.7.15 Electrical cable protection. The equipment design shall preclude damage to electrical cabling during all conditions of assembly, removal, insertion, and tilting of the equipment.

3.7.16 RF connectors. RF connectors shall exclude connectors using ferromagnetic materials and shall be selected from MIL-STD-242.

3.7.17 Mating connectors. A complete set of mating halves of connectors, backshells, and protective caps shall be provided. The mating connectors shall comply with the Improper application prevention paragraph of MIL-E-16400.

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3.7.18 Controls, indicators, and panel layout. The design of controls and indicators and their arrangement on operator and maintenance panels shall be in accordance with the Controls, indicators, and panel layout paragraph of MIL-E-16400. The abbreviations of MIL-STD-12 shall be used for equipment labeling where applicable.

3.7.19 Unacceptable materials. The equipment design shall comply with the Unacceptable materials paragraph of MIL-E-16400. The equipment design shall not include the materials specified in a through f:

a. Flammable or combustible materials
b. Polychlorinated biphenyls
c. Asbestos and asbestos compounds
d. Fragile or brittle materials
e. Beryllium and beryllium compounds unless so identified
f. Lithium and lithium compounds unless specifically approved by the procuring activity

3.7.20 Derating criteria. The derating criteria of NAVSEA 0967-LP-597-1011 shall be utilized in the design of the equipment.

3.7.21 Soldering. Soldering shall conform to the requirements of WS-6536E with SCN 1 and SCN 2.

3.7.22 Power supply design. Guidance for the design and construction of power supplies is provided in NAVMAT P4855-1. Power density in excess of 2 W per cubic inch shall require the approval of the Government. The junction temperature of semiconductor devices shall not exceed +110°C under worst case conditions. The power supply shall have a reserve capacity of at least 40 percent. Hot spot temperatures shall not exceed 40°C rise above ambient temperature with a maximum temperature of +110°C for parts less than or equal to 3 W of dissipation; 55°C rise above ambient temperature with a maximum temperature of +125°C for parts greater than 3 W of dissipation; 30°C rise above ambient temperature with a maximum temperature of +100°C for transformers; and 10°C rise above ambient temperature, due to self-heating, with a maximum temperature of +85°C for capacitors. Power supplies shall be designed to be maintainable at either intermediate or depot level; they shall not be limited by design to repair by the manufacturer. Power supplies shall not be encapsulated or embedded (potted) unless it can be shown by analyses and test to be necessary for heat removal or dissipation. This requirement shall not exclude conformal coating. The mean-time-between-failures requirement shall be not less than 40,000 hours at 55°C. Power supplies shall be designed to withstand and shall not be damaged by any load between an open-circuit and a short-circuit.

3.7.23 Magnetic materials. Non-magnetic materials shall be used for all parts except where magnetic materials are essential, in which case a listing of the size, quantity, and application of magnetic materials with a permeability greater than 2.0 shall be provided to the Government for approval prior to engineering release.

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3.7.24 Microelectronic devices. All microelectronic devices shall be in accordance with MIL-M-38510 and MIL-HDBK-175. Microprocessors with more than an 8-bit data bus or more than a 16-bit address bus shall not be employed in the design of the equipment without the written approval of the procuring agency. Module sockets for microelectronic devices shall not be employed.

3.7.25 Structural welding. Structural welding shall conform to MIL-STD-454, Requirement 13.

3.8 Identification and marking. Receiving set identification and marking shall conform to the requirements specified in 3.8.1 through 3.8.4.

3.8.1 Receiving set identification. Marking of all receiving sets shall be in accordance with the requirements of MIL-P-15024.

3.8.2 Visibility. When assembled, identification markings of parts shall be readily visible. Auxiliary markings, such as on access panels, shall be provided so as to identify those parts where labeling is not readily visible.

3.8.3 Readability of identification plates. Nomenclature plates and serial numbers shall be located so they are readable after receiving set installation. They shall be located on the front surface of the control panel of below-deck equipment and on the side of above-deck equipment.

3.8.4 Identification of wiring. Wiring shall be identified by color-coding, or by other means, such as marking of terminal at both ends of each lead. Color-coding for chassis wiring shall conform to the requirements of MIL-STD-681.

3.9 Prime power. Prime power to the combiner-demodulator and demultiplexer shall be 115 V or 230 V, single-phase, 50 Hz to 60 Hz. The entire receiving set performance shall be achieved when operated from power specified herein and in accordance with Type I of DoD-STD-1399, Section 300, and shall conform to the requirements and constraints thereof except that the maximum voltage shall be 115 V or 230 V.

3.9.1 Protection devices. Provision shall be made to automatically protect the receiving set when the voltage exceeds the limits specified in DoD-STD-1399, Section 300. Provision shall be made to automatically protect the DC power supply in the event of short-circuits or over-voltages. This provision for protection shall be extended to all pluggable units within the receiving set. A failure in one module shall not result in chain reaction failures in other modules. Open-circuits and short-circuits in interface cabling shall not result in damage.

3.9.2 Prime power control. Prime power circuit breakers shall be located on the combiner-demodulator and on the demultiplexer. A status indication shall be provided on the circuit breakers.

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3.10 Software design. The software development design practices of DoD-STD-2167 shall be adhered to in the development of computer software for the microprocessor(s) of the equipment. All software programs shall be ROM-resident and executed. The requirements specified in 3.10.1 and 3.10.2 shall be adhered to.

3.10.1 Microprocessor resource allocation and reserves. Memory storage, input and output channels, processing time requirements, and throughput for each microprocessor used in the system shall be determined. Total system memory reserves of at least 100 percent and processing time reserves of at least 50 percent shall exist at the time of equipment acceptance by the procuring activity. These reserves shall be in addition to the capacities required for all known and planned processing.

3.10.2 Software regeneration and maintenance. Deliverable code shall be in accordance with the Regenerable and maintainable code paragraph of DoD-STD-2167.

3.11 Workmanship. Workmanship shall be as specified in 3.11.1 and 3.11.2.

3.11.1 General workmanship. Workmanship shall conform to Requirement 9 of MIL-STD-454.

3.11.2 Workmanship screen. All equipment produced in accordance with this specification shall withstand a random vibration screen and a temperature screen.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to the requirements of section 3 and section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either

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indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality and safety assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality and safety program are being properly applied, b) Government product inspection to measure quality and safety of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
    3. Environmental inspection (see 4.4.3)
c. Inspection of preparation for delivery (see 4.11)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV and TABLE V.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in 4.4.1 through 4.4.5.

4.4.1 Production inspection (Group A). Production inspection shall be made on every receiving set offered for delivery. The inspection shall comprise such examination and testing which will prove the workmanship and reveal the omissions and errors of the production process such as, functional and performance tests at a limited number of points, and tests which detect hidden defects of material. Production inspection shall consist of the examinations and tests specified in TABLE IV and TABLE V, Group A.

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TABLE IV. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.2.2.5, 3.2.3.9, 3.2.4.14, 3.2.5.9	4.5.1	X	X		
Size	3.2.2.9, 3.2.3.12, 3.2.4.17, 3.2.5.12	4.5.1	X	X		
Parts, materials, and processes	3.6 through 3.6.7	4.5.1	X	X		
Finish	3.7.14	4.5.1	X	X		
Marking	3.7.3.3, 3.8.1 through 3.8.4	4.5.1	X	X		
Safety	3.7.3, 3.7.3.1, 3.7.3.5 through 3.7.3.6.1	4.5.1	X	X		
Design and construction	3.7.1, 3.7.2, 3.7.4 through 3.7.8.1, 3.7.10 through 3.7.13, 3.7.15 through 3.7.20 3.7.22 through 3.7.24	4.5.1	X	X		
Environmental requirements						
Temperature	3.5.1.1, 3.5.2.1	4.5.4	X			X
Salt fog	3.5.1.2, 3.5.2.2	4.5.6	X			X
Altitude	3.5.1.3, 3.5.2.3	4.5.11	X			X
Wind	3.5.1.4	4.5.7	X			X
Ice loading	3.5.1.5	4.5.8	X			X
Sea state	3.5.1.6, 3.5.2.4	4.5.13	X			
Shock	3.5.1.7, 3.5.2.5	4.5.14	X			X
Vibration	3.5.1.8, 3.5.2.6	4.5.15	X			X

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TABLE IV. Examinations and tests. - (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Electromagnetic compatibility	3.5.1.9, 3.5.2.7	4.5.9	X			X
Enclosure	3.2.4.15, 3.2.5.10, 3.5.1.10, 3.5.2.8	4.5.16	X		X	
Humidity	3.5.1.11, 3.5.2.9	4.5.5	X			X
Inclination	3.5.1.12, 3.5.2.10	4.5.17	X			X
EMP	3.5.1.13, 3.5.2.11	4.5.9.1	X			
Nuclear effects	3.5.1.14 through 3.5.1.14.3, 3.5.2.12	4.5.10	X			
Fungus	3.6.8	4.5.18	X			
Acoustic noise	3.7.3.2	4.5.19	X			
Interchangeability	3.7.2.1	4.10	X			
Leakage current	3.7.3.4	4.5.12 through 4.5.12.2	X			
Thermal design	3.7.9, 3.7.9.1	4.5.1	X		X	
Reliability	3.3	4.8 through 4.8.3	X			
Maintainability	3.4	4.9 through 4.9.5	X			
Welding	3.7.25	4.5.1	X		X	
Soldering	3.7.21	4.5.1	X		X	
Prime power	3.9 through 3.9.2	4.5.1	X		X	
Software design	3.10	4.6 through 4.6.1.7.5	X			
Microprocessor resource allocation and reserves	3.10.1	4.6 through 4.6.1.7.5	X			
Software regeneration and maintenance	3.10.2	4.6 through 4.6.1.7.5	X			
Workmanship	3.11.1	4.5.1	X	X		
Workmanship screen	3.11.2	4.7	X	X		
Operating test		4.5.2	X	X		
Pre- and post-test functional verification		4.5.3	X			X

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TABLE V. Performance tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Receiving set						
Function	3.1.2.1 through 3.1.2.4	4.5.1	X			
Frequency of operation	3.2.1.1	4.5.1	X	X		
C/N ratio	3.2.1.2	4.5.1	X	X		
C/N enhancement	3.2.1.3	4.5.1	X	X		
Receiving set bandwidth	3.2.1.4	4.5.1	X	X		
BER with PSK modulation	3.2.1.5	4.5.1	X	X		
Interconnecting cables	3.2.1.6	4.5.1	X			
Broadband radiated noise	3.2.1.7	4.5.1	X			
Spurious responses	3.2.1.8	4.5.1	X	X		
Half-IF response	3.2.1.9	4.5.1	X	X		
Image response	3.2.1.10	4.5.1	X	X		
IF rejection	3.2.1.11	4.5.1	X	X		
Desensitization	3.2.1.12	4.5.1	X	X		
Circuit test points	3.2.1.13	4.5.1	X			
Noise environment	3.1.3.1	4.5.1	X	X		
Antenna						
Frequency	3.2.2.1	4.5.1	X			
VSWR	3.2.2.2	4.5.1	X	X		
Polarization	3.2.2.3	4.5.1	X	X		
Antenna pattern	3.2.2.4	4.5.1	X			
Antenna mounting	3.2.2.6	4.5.1	X			
Antenna feedlines	3.2.2.7	4.5.1	X			
Antenna coaxial cable	3.2.2.8	4.5.1	X			
Amplifier-converter						
Impedances	3.2.3.1	4.5.1	X		X	
Frequencies	3.2.3.2	4.5.1	X	X		
Signal levels	3.2.3.3	4.5.1	X	X		
Preselector filter	3.2.3.4	4.5.1	X		X	
Isolation	3.2.3.5	4.5.1	X		X	
Input termination	3.2.3.6	4.5.1	X		X	
Input protection	3.2.3.7	4.5.1	X		X	
Preamplifier	3.2.3.8	4.5.1	X			
Amplifier-converter mounting	3.2.3.10	4.5.1	X			
Amplifier-converter packaging	3.2.3.11	4.5.1	X	X		

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TABLE V. Performance tests. - (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Combiner-demodulator						
Impedances	3.2.4.1	4.5.1	X		X	
Isolation	3.2.4.2	4.5.1	X		X	
Interface ports signal characteristics	3.2.4.3.1, 3.2.4.3.2, 3.2.4.3.3, 3.2.4.3.4	4.5.1	X	X		
		4.5.1	X			
		4.5.1	X	X		
		4.5.1	X	X		
LO characteristics	3.2.4.4	4.5.1	X	X		
Channel units	3.2.4.5	4.5.1	X	X		
Pulse blanking	3.2.4.6, 3.2.4.6.1	4.5.1	X	X		
In-band overload protection	3.2.4.7	4.5.1	X	X		
Signal combiner	3.2.4.8	4.5.1	X	X		
Demodulation unit	3.2.4.9 through 3.2.4.9.11	4.5.1	X	X		
Amplifier-converter DC power control	3.2.4.10	4.5.1	X	X		
Signal monitor	3.2.4.11	4.5.1	X	X		
Fault indicators	3.2.4.12	4.5.1	X	X		
Elapsed time meter	3.2.4.13	4.5.1	X			
Drawer harnesses and cables	3.2.4.16	4.5.1	X			
Demultiplexer						
Input impedance	3.2.5.1	4.5.1	X		X	
Input level	3.2.5.2	4.5.1	X	X		
Bit integrity	3.2.5.3	4.5.1	X	X		
Output circuitry	3.2.5.4 through 3.2.5.4.5	4.5.1	X	X		
Signal monitor	3.2.5.5	4.5.1	X			
Fault indicators	3.2.5.6	4.5.1	X	X		
Status indicators	3.2.5.7	4.5.1	X	X		
Elapsed time meter	3.2.5.8	4.5.1	X			
Drawer harnesses and cables	3.2.5.11	4.5.1	X			
Receiving set interfaces	3.2.6 through 3.2.6.4	4.5.1	X		X	

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4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis, where sample size is determined using special inspection level S-3 as specified in MIL-STD-105 for normal, tightened, and reduced inspection. Failure of any unit within the sample to conform to prescribed requirements shall constitute a failure and shall be cause for rejection of the inspection lot. Production control inspection shall include those examinations and tests specified in TABLE IV and TABLE V, Group B and shall be performed on receiving sets that have passed production inspection. The receiving sets shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identify.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. Environmental inspection shall consist of the examinations and tests specified in TABLE IV, Group C, and shall be performed on receiving sets which have been subjected to, and have passed, production inspection. An operating test shall be performed during the various environmental conditions for monitoring receiving set operation. After the receiving sets have been subjected to all the specified environmental conditions, the performance tests of TABLE V shall be performed on the receiving sets.

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected without regard to their quality. One sample shall be selected from the first 20 production units and one sample shall be selected from each successive 100 production units, or fraction thereof, produced. The first article inspection may be used to satisfy this requirement for the first 20 production units.

4.4.4 Nonconformance corrective action and retest. If a sample unit fails the inspection specified in 4.4.2 or 4.4.3, the contractor shall immediately investigate the cause of the failure and shall implement the corrective action necessary to correct the units of product which were manufactured under the same conditions, materials, processes, and so forth. The receiving sets shall then be retested.

4.4.5 Reinspection of production control and environmental test units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to, and have passed production control and environmental inspection may be accepted on the contract provided they are resubjected to, and pass, production inspection specified in 4.4.1 after repair of all visible damage.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.19.

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4.5.1 Examinations and tests. Examinations and tests specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400 and Government-approved test procedures. It shall be the responsibility of the contractor to provide all cables, connectors, test instrumentation, test facilities, and other devices required to set up and perform all tests at the contractor's plant, or other place of test, to verify compliance with the requirements specified herein. When the receiving set is tested as a system, the contractor shall provide interconnecting cables adequate to simulate realistic installation conditions; that is, length, attenuation, and shielding shall be as identical as possible to actual installations.

4.5.2 Operating test. The receiving set shall be energized and subjected to an operating test to ensure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to the requirements specified herein.

4.5.3 Pre- and post-test functional verification. Prior to proceeding with any of the Group C test methods, the test item shall be operated under standard ambient conditions, and a record shall be made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking satisfactory performance of the test item either during or at the conclusion of the test, or both, as required. When operation of the test item is required during the test exposure, the operation and performance checks shall be of sufficient duration or shall be repeated at appropriate intervals, to ensure a record of comprehensive comparative data for comparison with data recorded under standard ambient conditions. The test item shall be visually examined, and a record shall be made of any damage or deterioration resulting from the test. If a test chamber is used for the test, a visual examination of the test item within the chamber shall be performed at test conditions, when possible. Upon completion of the test, a visual examination of the test item shall be performed again after the test item has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from conforming to operational service or maintenance requirements, shall provide reason to consider the test item as having failed to withstand the conditions of the test. Leakage or degradation of impregnating compounds shall be considered damage and cause for rejection.

4.5.4 Temperature tests. The equipment shall be tested in accordance with MIL-STD-810, Method 501.2, Procedure I and Procedure II, and Method 502.2, Procedure I and Procedure II, to demonstrate conformance with 3.5.1.1 and 3.5.2.1.

4.5.5 Humidity test. The equipment shall be tested in accordance with MIL-STD-810, Method 507.2, Procedure 1, Cycle 1 to demonstrate conformance with 3.5.1.11 and 3.5.2.9.

4.5.6 Salt fog test. The equipment shall be tested in accordance with MIL-STD-810, Method 509.2 to demonstrate conformance with 3.5.1.2 and 3.5.2.2.

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4.5.7 Wind speed. The above-deck equipment shall be tested to demonstrate conformance with 3.5.1.4.

4.5.8 Ice loading. The above-deck equipment shall be tested in accordance with MIL-STD-810, Method 521.0 to demonstrate conformance with 3.5.1.5.

4.5.9 Electromagnetic environment effects (E^3) (including EMP). The equipment shall be tested in accordance with MIL-STD-462 to demonstrate conformance with 3.5.1.9 and 3.5.2.7.

4.5.9.1 EMP. The equipment shall be tested in accordance with MIL-STD-462 to demonstrate conformance with 3.5.1.13 and 3.5.2.11.

4.5.10 Nuclear effects testing. Verification of conformance of hardness to nuclear effects shall be demonstrated either by analysis or nuclear simulation tests to show compliance with 3.5.1.14 and 3.5.2.12. Electronic piece-parts and circuits for which TREE test data are available on the specific device can be verified by analysis. Test Plans submitted to the Government shall include, a) Facility to be used, b) Parts, circuits, or equipments to be tested, c) Levels of test, d) Diagnostic techniques, e) Description of data reduction methodology. For direct tests of shock, blast, or thermal effects, test plans shall include the test waveforms, interface circuits to be tested, test equipment, test fixtures, diagnostic techniques, and description of data reduction.

4.5.11 Altitude. The equipment shall be subjected to low-pressure testing in accordance with MIL-STD-810, Method 500.2, Procedure I and Procedure II, to demonstrate conformance with 3.5.1.3 and 3.5.2.3.

4.5.12 Leakage current test. Leakage current shall be measured at 115 V and 230 V, 50 Hz to 60 Hz, single-phase source power.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRICAL SHOCK PROTECTION.

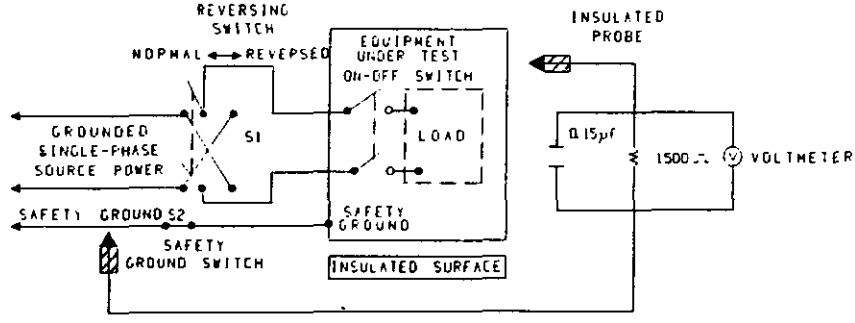
THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.12.1 Receiving set connections. After power removal, each receiving set directly connected to an external power source and units deriving power from the receiving set shall be placed on an insulated surface. All safety ground conductors between the receiving set and units deriving power from the receiving set shall be intact. The safety ground conductor between the receiving set under test and the source power shall be opened during the test. OBSERVE WARNING STATEMENT. The receiving set shall be connected as specified in FIGURE 2.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED. ON-OFF SW ON. REPEAT STEP 3.
7. ON-OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT.



WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 2. Single-phase test diagram for leakage current measurement.

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4.5.12.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 mA of leakage current. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 2. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.13 Sea state. An analysis shall be conducted to demonstrate conformance with 3.5.1.6 and 3.5.2.4.

4.5.14 Shock test. The equipment shall be tested in accordance with MIL-S-901, Grade A, Class 1, Type A to demonstrate conformance with 3.5.1.7 and 3.5.2.5. Any weld that is broken by shock testing shall be cause for rejection. Inoperability of the equipment during and after each blow of the shock test machine is allowable only if the system is self-correcting, requiring no operator interface. Malfunction requiring operator actions (that is, physical reset of circuit breakers, switches, and so forth) shall be cause for rejection.

4.5.15 Vibration test. The equipment shall be tested in accordance with MIL-STD-167-1, Type I without internal or external resilient mountings to demonstrate conformance with 3.5.1.8 and 3.5.2.6.

4.5.16 Enclosure test. The equipment shall be tested in accordance with MIL-STD-108 to demonstrate conformance with 3.5.1.10 and 3.5.2.8.

4.5.17 Inclination test. The equipment shall be tested in accordance with the Inclination test paragraph of MIL-E-16400 to demonstrate conformance with 3.5.1.12 and 3.5.2.10.

4.5.18 Fungus. The receiving sets shall be subjected to the fungus test specified in Method 508.3 of MIL-STD-810 to demonstrate conformance with 3.6.7.

4.5.19 Acoustic noise. The equipment shall be tested in accordance with MIL-STD-740-1 to demonstrate conformance with 3.7.3.2.

4.6 Software quality assurance provisions. Tests shall be performed to ensure that the software conforms to all specified technical, performance, and operational requirements and acceptance criteria. Testing shall be performed in accordance with the requirements of the Coding and Unit Testing, CSC Integration and Testing, and CSCI Testing paragraphs of DoD-STD-2167, unless otherwise specified in 4.6.1 through 4.6.1.7.5.

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4.6.1 Testing levels and requirements. Software testing shall be accomplished at the levels specified in a through g:

a. Coding and unit tests (informal)
b. Lower-level computer software component (LLCSC) tests (informal)
c. Top-level computer software component (TLCSC) tests (informal)
d. Computer Software Configuration Item tests (CSCI) (formal)
e. System integration tests (informal)
f. System performance tests (formal)
g. Stress testing (formal)

Informal testing shall conform to the requirements specified in h and i:

h. Tests shall be monitored by the developer and shall be subject to informal monitoring by the Government or its representative.
i. The tests shall constitute internal milestones and informal project milestones.

Formal testing shall conform to the requirements specified in j through m:

j. The test shall constitute an official project milestone.
k. The test shall be officially witnessed by the Government during its performance and shall be conducted in accordance with previously approved test specifications and procedures.
l. All items that affect the test or are used in the test, including hardware and software, must be certified before the test.
m. The test shall be subsequently audited and reviewed by the Government representative.

4.6.1.1 Coding and unit tests (informal). Each unit, the smallest logical entity, shall undergo the tests specified in a through d as a minimum:

a. Peer review
b. Error-free compilation
c. Exercise of logical execution paths
d. Validation of intended function

Upon completion of unit testing, the unit shall be placed under the software configuration management of the developer. The tests shall be conducted as specified in the Coding and Unit Testing paragraph of DoD-STD-2167.

4.6.1.2 LLCSC tests (informal). Each LLCSC shall be tested in accordance with the CSC Integration and Testing paragraph of DoD-STD-2167.

4.6.1.3 TLCSC tests (informal). Each TLCSC shall be tested in accordance with the CSC Integration and Testing paragraph of DoD-STD-2167.

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4.6.1.4 CSCI tests (formal). Each CSCI shall be tested in accordance with the CSCI Testing paragraph of DoD-STD-2167. The tests are considered preliminary software acceptance tests. Government-approved test specifications, test plans, and test procedures shall be used to conduct this level of testing.

4.6.1.5 System integration test (informal). The system integration test shall encompass and validate system integration of software with hardware, with other software and with other subsystems. The system integration test shall serve as a further check on man-machine interface, functional completeness, and performance requirements. It shall involve the testing of software-software and software-hardware interfaces either as subsystems are integrated into a larger system, or as one system is integrated with another. The contractor shall plan for and demonstrate progress against this plan to the Government during system integration testing. Specific integration milestones shall be identified and scheduled.

4.6.1.6 System performance test (formal). The contractor shall conduct a formal system performance test, which is the final verification test to certify that the system performs as specified herein and in the contract. The contractor shall schedule, and the Government shall witness, a system performance test to certify that the hardware and software representing the system have met all performance requirements. As a minimum, software system performance testing shall be designed to:

a. Verify the total man-machine interface
b. Validate system initiation, data entries, and the monitoring and control of system operation
c. Verify the interfacing of equipment identified in this specification
d. Verify the capability of the software to satisfy applicable systems, program performance, and quality assurance requirements
e. Verify inter-system and intra-system formats and interfaces

4.6.1.7 Stress testing (formal). Stress testing shall be carried out on the system as specified in 4.6.1.7.1 through 4.6.1.7.5. This test shall verify the capability of the system to properly handle and survive erroneous inputs. Government-approved test specifications and procedures shall be used to conduct this level of testing.

4.6.1.7.1 Stress testing error priorities. Stress testing errors shall be prioritized by severity as specified in a through e:

a. Priority 1. An error which prevents the accomplishment of an operational or mission essential function in accordance with official requirements (e.g., causes a program stop), which prevents the operator's accomplishment of an operational or mission essential function, or which jeopardizes personnel safety

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b. Priority 2. An error which adversely affects the accomplishment of an operational or mission essential function in accordance with official requirements so as to degrade performance and for which no alternative work-around solution exists, or which adversely affects the operator's accomplishment of an operational or mission essential function so as to degrade performance and for which no alternative work-around solution exists (Reloading or restarting the software is not an acceptable work-around solution.)

c. Priority 3. An error which adversely affects the accomplishment of an operational or mission essential function in accordance with official requirements so as to degrade performance and for which there is reasonable alternative work-around solution, or which adversely affects the operator's accomplishment of an operational or mission essential function so as to degrade performance and for which there is a reasonable alternative work-around solution (Reloading or restarting the software is not an acceptable work-around solution.)

d. Priority 4. An error which is an operator inconvenience or annoyance and does not effect a required operational or mission essential function

e. Priority 5. All other errors

4.6.1.7.2 Errors during tests. Occurrence of a software error of a priority one or two severity during the software stress test shall require correcting the error and repeating the test in its entirety. Occurrence of an intermittent software error of a priority one or two severity shall require that the particular operation(s) be repeated several times to the satisfaction of the contracting activity, and that the full stress test then be repeated in its entirety. There shall be no patching to correct errors while the stress test is in progress. The only other patches that may be in the software during the software stress test are those which are considered to be required for testing purposes and they shall be specifically set forth in the test procedures. Caution should be exercised that these patches do not interfere with the validity of the software stress test results.

4.6.1.7.3 Stress test limitations. Error and patch limits relevant to the software undergoing stress testing are specified in 4.6.1.7.4 and 4.6.1.7.5. These limits shall be satisfied prior to the commencement of the test. Errors detected during the test which cause this criteria to be exceeded shall invalidate the test. An individual test may continue to run to its planned completion in order to uncover any additional errors. In those tests where the error limits are exceeded, the test shall be rerun in its entirety. In those tests where the patch limits are exceeded, consequently requiring source level changes and recompilation or assemblage, the test shall be rerun in its entirety.

4.6.1.7.4 Error limits for software acceptance. The error limits specified in a through c shall apply at the time of software acceptance by the contracting activity.

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a. The number of unresolved software errors (excluding documentation errors) shall not exceed the specifications of TABLE VI.

TABLE VI. Software stress test error limits.

Severity	Limits
Priority 1 and 2 (high)	Zero
Priority 3 (medium)	One per 70K of machine instruction words or fraction thereof
Priority 4 and 5 (low)	One per 35K of machine instruction words or fraction thereof

b. Intermittent software errors shall be included in the count of software errors and receive no special consideration.

c. The number of unresolved technical errors in all of the deliverable documentation shall not exceed the sum of three, plus one for every 25K of machine instructions or fraction thereof.

4.6.1.7.5 Patch limits for software acceptance. The patch limits specified in a and b shall apply at the time of software acceptance by the contracting activity.

a. The total number of patch words in the software shall not exceed 0.005 times the total machine instruction words in the software.

b. All patches shall be documented.

4.7 Workmanship screen. The receiving set shall be subjected to a random vibration screen and temperature screen as specified in 4.7.1 and 4.7.2 to demonstrate conformance with 3.11.2.

4.7.1 Random vibration screen. Prior to conducting temperature cycling, vibration shall be performed on all receiving sets. The receiving sets shall be mounted in their normal operational position (without shock isolators) and the vibration stress shall be applied vertical to this position for an accumulated time of 10 minutes. The vibration shall be in accordance with FIGURE 3. The control accelerometer shall be located next to one of the mounting points of the item under test. The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz, and a 100 Hz bandwidth above 500 Hz. The test item shall be energized during vibration and an appropriate input signal shall be applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during the test shall require corrective action before the test is resumed.

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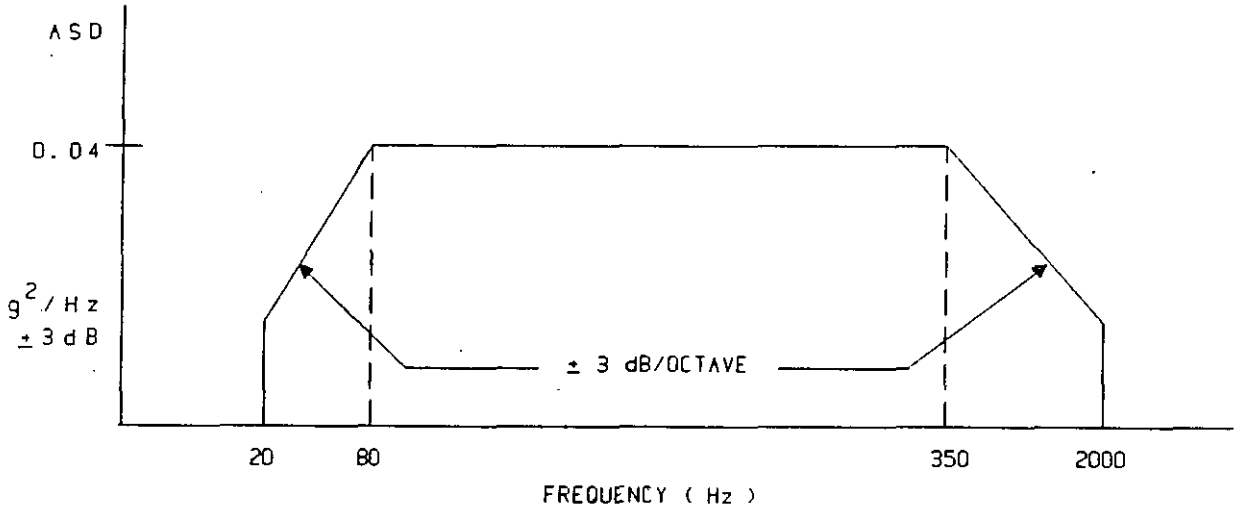


FIGURE 3. Random vibration.

4.7.2 Temperature screen. Each receiving set shall be subjected to 10 cycles of the temperature curve as specified in FIGURE 4. The temperature rate of change shall be not less than 5°C per minute. Receiving set power shall be turned on and off at the indicated times. Performance measurements on the antenna and amplifier-converter shall be carried out at the above-deck temperatures specified in FIGURE 4. Performance measurements on the combiner-demodulator and demultiplexer shall be carried out at the below-deck temperatures specified in FIGURE 4. The receiving set shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the receiving set reaches 80 percent of the chamber temperature. When failures occur, the receiving set shall be reworked and the cycling shall be continued for a cumulative total of 10 cycles.

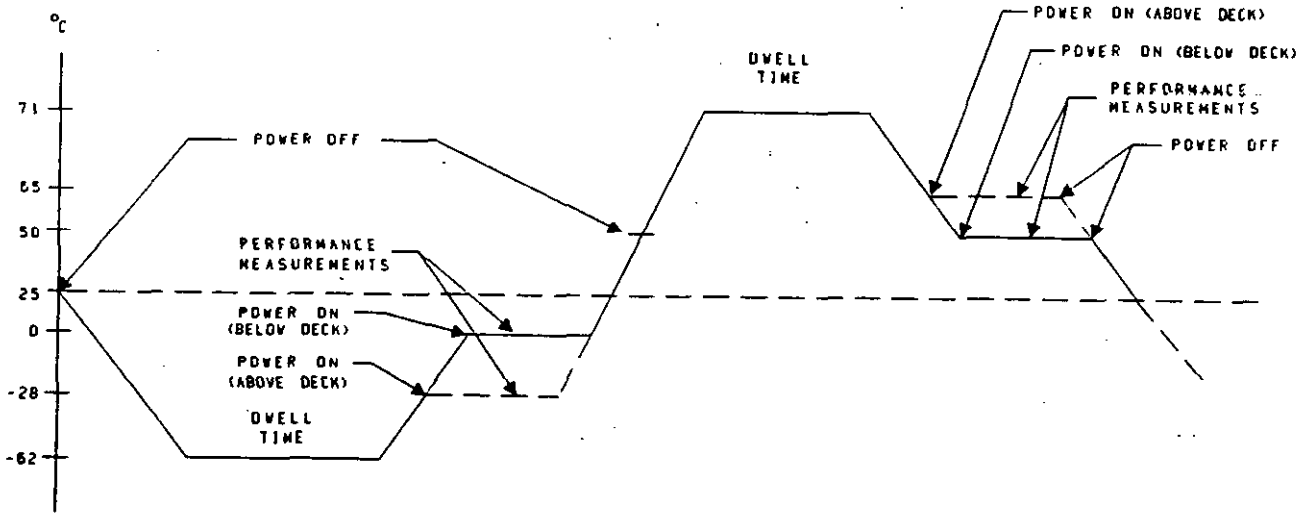


FIGURE 4. One cycle of temperature curve (not to scale).

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4.8 Reliability test. Conformance to the reliability requirements of 3.3 shall be verified by the demonstration specified in 4.8.1 through 4.8.3.

4.8.1 Reliability demonstration. A production reliability acceptance test shall be performed on one of the first five production receiving sets in accordance with the general requirements and Task 302 of MIL-STD-781 and MIL-HDBK-781 for category 3A equipment. Testing shall proceed to a decision in accordance with criteria specified for category 3A equipment, Test Plan VIII-D of MIL-HDBK-781. This testing shall be completed to an accept decision before any production receiving sets can be accepted.

4.8.2 Relevant failure. The final decision as to the relevancy or nonrelevancy of a failure shall be made by the procuring activity using the Failure categories paragraph of MIL-STD-781 as a guide. Any operating discrepancy or degradation that required an unscheduled adjustment or calibration to be made, after initial satisfactory operation of the receiving sets under test, shall be considered a relevant failure.

4.8.3 Rejected receiving sets. When a receiving set has been rejected as a result of a reliability test or when pattern failures occur, the entire lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test in accordance with 4.8.1. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on all receiving sets delivered under the contract.

4.9 Maintainability demonstration. The receiving set maintainability requirements of 3.4 shall be demonstrated in accordance with Method 4 of MIL-STD-471 by the replacement of subassemblies (modules, PC cards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. The contractor shall determine 50 candidate faults for the demonstration. A representative of the procuring activity shall select a sample of 20 faults from these candidate faults to be used in this part of the demonstration. For fault simulation, a number shall be assigned to each component or part within a subassembly or chassis-mounted component or part. Fault simulation for modules shall be accomplished by introduction of faulty parts corresponding to the number selected.

4.9.1 Demonstration environment. The demonstration at the organizational level shall be performed in a space and temperature environment similar to that in which the receiving set will be installed.

4.9.2 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the equipment maintainability demonstration.

4.9.3 Technical documentation. Technical documentation to be used in the demonstration shall be limited to the technical manual.

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4.9.4 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government, a) Develop an approach for redesign or correction of all deficiencies, b) Upon procuring-activity approval of an approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.9.5 Acceptance. The maintainability demonstration shall exhibit that the receiving sets conform to the accept criteria prior to delivery of any equipment.

4.10 Interchangeability test. The contractor shall perform interchangeability tests to verify conformance with the requirements of 3.7.2.1. The test configuration shall enable RF injection of a Quick Brown Fox message into the AN/SSR-1A antenna to simulate reception and processing of an actual satellite signal. It shall be demonstrated that the receiving set will reproduce the Quick Brown Fox message. The receiving set shall also pass a test where receiving set modules supplied by the Government are substituted into the contractor's unit and proper performance of the receiving set is shown to continue without impediment. The contractor's unit shall continue to reproduce the Quick Brown Fox message regardless of which or how many Government modules are substituted for contractor modules. This module substitution test shall also be performed for substitution of contractor modules into Government receiving test set(s). These tests shall be performed prior to delivery of the first production unit and shall be performed at the contractor's plant.

4.11 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The receiving set covered by this specification is intended for receiving TTY broadcasts that originate at a ground station and are relayed by a satellite. The transmitted carrier may contain either FM or PSK modulation. A switch within the receiving set provides for selecting the

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applicable demodulation mode. In the receiving set, the signal is processed to derive outputs for operating TTY equipment. The receiving set is capable of driving either high-level or low-level interfaces.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Use of other than current documents listed in the DoDISS (see 2.1.1)
c. Type of TTY switch drivers (high-level or low-level) required
d. Number of first article samples to be submitted if other than specified in 4.3
e. When reinspected production control and environmental inspection samples may not be accepted (see 4.4.5)
f. Applicable levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.3 and 4.4	Inspection and Test Reports	DI-T-5329
4.4	Procedures, Test	UDI-T-22710B

(Copies of DIDs required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of DAR 7-104.55. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Installation responsibility. The Government will select antenna locations and equipment installations.

Preparing activity:
NAVY - EC

(Project 5820-N861(EC))

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APPENDIX

AN/SSR-1A SYNTHESIZED LOCAL OSCILLATOR (SLO) MODULE

10. SCOPE

10.1 Scope. This appendix covers the specifications for the SLO module.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the DoDISS specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic, Interior Communication, And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-S-19500	Semiconductor Devices, General Specification For
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification For

STANDARDS

MILITARY

MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I-Environmental And Type II-Internally Excited)
MIL-STD-188-114	Electrical Characteristics Of Digital Interface Circuits
MIL-STD-198	Capacitors, Selection And Use Of
MIL-STD-199	Resistors, Selection And Use Of
MIL-STD-242	Electronic Equipment Parts Selected Standards, Connectors
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 070	Interface Standard For Shipboard Systems, D.C. Magnetic Field Environment

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APPENDIX

HANDBOOKS

MILITARY

MIL-HDBK-217 Reliability Prediction Of Electronic Equipment

30. REQUIREMENTS

30.1 General. Except as otherwise specified herein, the SLO module shall be produced such that the form, fit, and function of the SLO is identical to that of the local oscillators that it replaces. The SLO module shall conform to the requirements of section 3 of this specification except as otherwise specified in this APPENDIX. SLO electrical and mechanical interfaces shall ensure backward compatibility with fleet AN/SSR-1A units and interchangeability with F003, F004, and F005 LO modules.

30.2 Electrical requirements. The SLO shall, as a minimum, be compatible with the existing interface requirements of the AN/SSR-1A system and shall, when replacing the existing local oscillator module, produce no degradation in BER. The SLO shall have the same interfaces as the existing module and in addition shall have a full duplex serial control port, 3 external fault inputs and an internal fault output. The SLO shall produce the same RF output level as the existing module.

30.2.1 Frequency. The SLO output shall cover the frequency band from 222.050 MHz to 252.045 MHz.

30.2.1.1 Step sizes. The SLO step size shall be 5 kHz.

30.2.1.2 Frequency control. Changing frequencies shall be accomplished by module-mounted switches in the local mode or a serial data stream in the remote mode. Local or remote mode shall be selectable by a switch located on the module edge.

30.2.1.3 Frequency change speed. In the remote mode, the frequency shall change and settle to less than 10° phase error from final phase within 100 milliseconds of an execute command.

30.2.1.4 Frequency temperature stability. The frequency stability of the output signal over the required temperature range shall be better than plus or minus 2 parts per million.

30.2.1.5 Frequency drift (aging). The output frequency shall not drift more than 2 parts per million per year.

30.2.1.6 Frequency stability (short-term). Short-term stability of the output frequency shall be within ± 1 part per 10^8 per second.

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30.2.2 RF output. The RF output level of the SLO module shall be +27 dBm as measured at the RF output connector.

30.2.2.1 Spurious and harmonics. Harmonics of the output signal shall be suppressed at least 10 dB below the desired output signal level. Spurious responses shall be at least 40 dB below the desired output signal level.

30.2.2.2 Phase noise. The phase noise shall be at least 75 dB per Hz down when measured 2 kHz from the carrier. However, in no case shall the noise be great enough to cause any increase in the BER of an AN/SSR-1A receiver. This requirement shall be met under the vibration requirements of MIL-STD-167-1, Type I.

30.2.3 Electrical interfaces. The electrical interfaces for the SLO other than the RF output shall be input power, serial remote control, fault inputs, and SLO fault output.

30.2.3.1 Input power. The input power to the SLO shall be +20 VDC with a nominal tolerance of ±7 percent. The SLO shall operate normally with the input power having ±25 percent voltage excursions lasting up to 3 seconds. The maximum allowed SLO power consumption shall be as listed below:

Voltage	Current
20	350 mA
-20	160 mA

30.2.3.2 Serial remote control. The SLO shall have a full duplex, asynchronous, serial, 600 bps remote control interface that meets the requirements of MIL-STD-188-114 for unbalanced operation. Each asynchronous character shall consist of a start bit, 7-bit American Standard Communication Information Interchange (ASCII) data field, an even parity bit, and 1 stop bit. The character transmission order shall be start bit, LSB through MSB, parity bit, and stop bit. The remote control interface shall respond to received commands and shall not initiate a transmission on its own. There are three basic command types that the SLO's remote control shall respond to, Frequency and Status, Reset Fault Hold, and Hardware Restart. For each command issued, the SLO shall respond with status information. This status information shall consist of operating frequency, local or remote control, fault and fault reset, with the actual message depending on what command was received, and the SLO status at the time. The command and response syntax and message content shall be as specified in 30.2.3.2.1 through 30.2.3.2.9.

30.2.3.2.1 Syntax. The syntax of the Frequency and Status command shall be:

FREQ_XXX.XXX(cr)(lf)

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The message is sent, reading left to right, where the XXX.XXX is the desired operating frequency of the radio in MHz. Only values in the range of 242.000 MHz to 271.995 MHz in 5 kHz steps are allowed. Note that the SLO operating frequency is related to the radio operating frequency by a 19.95 MHz offset; that is, when the requested radio operating frequency is 242.000 MHz the SLO operating frequency would be 222.050 MHz. The space (_) and decimal point (.) are required. The (cr) is the ASCII character carriage return and the (lf) is the ASCII character line feed.

30.2.3.2.2 Parity error. If the SLO detects a parity error in the message the SLO shall reject the command and respond with a parity error message. This message syntax shall be:

ERROR_PARITY(cr)(lf)

END(cr)(lf)

30.2.3.2.3 Invalid command. If the command is invalid, such as a misspelling, wrong command, or frequency out of range, the command shall be rejected and the response shall be:

ERROR_COMMAND(cr)(lf)

END(cr)(lf)

30.2.3.2.4 Local control. If the SLO is in local control, the command shall be rejected and the response shall be:

FREQ_XXX.XXX(cr)(lf)

MODE_INT(cr)(lf)

FAULT_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
(optional entries)
FAULT_HOLD_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
(optional entries)

END(cr)(lf)

Where the XXX.XXX is the SLO operating frequency as set locally. The MODE_INT indicates local operation. The FAULT message shall indicate faults that are present at the time the message is sent, whereas FAULT_HOLD shall indicate faults that are not now present but have previously been detected. Multiple faults shall be sent in the same message with each fault character(s) separated by a comma and no spaces. The fault fields shall be sent with each transmission but the fault character(s) are to be present only when a fault has been detected and are shown as optional entries in the message. The fault codes shall be:

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APPENDIX

Code	Explanation
A	External fault input 'A'
B	External fault input 'B'
C	External fault input 'C'
SHF	SLO high frequency loop out of lock
SLF	SLO low frequency loop out of lock
SPWR	SLO output power is low
SMCP	SLO microcomputer fault (e.g., memory checksum error)

30.2.3.2.5 Correct command. If the SLO is in the remote control mode and a valid frequency command is received, i.e., within the 242.000 MHz to 271.995 MHz range in 5 kHz increments, the response shall be:

FREQ_XXX.XXX(cr)(lf)
MODE_EXT(cr)(lf)
FAULT_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
(optional entries)
FAULT_HOLD_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
(optional entries)
END(cr)(lf)

30.2.3.2.6 Frequency command error. If the frequency command is not valid, the frequency shall be ignored and the response shall be:

FREQ_ERROR(cr)(lf)
MODE_EXT(cr)(lf)
FAULT_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
(optional entries)
FAULT_HOLD_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
(optional entries)
END(cr)(lf)

30.2.3.2.7 Reset fault hold. The Reset Fault Hold command shall cause the latched faults to be cleared to begin a new accumulation period. The command is:

RESET(cr)(lf)

The response shall be:

FAULT_CLEAR(cr)(lf)
END(cr)(lf)

30.2.3.2.8 Combined commands. The Reset Fault Hold command shall be acceptable when combined with the Frequency and Status command as long as the frequency portion of the command is sent first. This combined command is:

FREQ_XXX.XXX,FAULT_CLEAR(cr)(lf)

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The response to the combined command, assuming no command error, shall be:

```
FREQ XXX.XXX(cr)(lf)
MODE EXT(cr)(lf)
FAULT_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
      (  optional entries   )
FAULT_HOLD_A,B,C,SHF,SLF,SPWR,SMCP(cr)(lf)
            (  optional entries   )
FAULT_CLEAR(cr)(lf)
END(cr)(lf)
```

30.2.3.2.9 Hardware restart. The Hardware Restart command shall cause the microcomputer in the SLO to begin an initialization routine which shall perform self test functions, program the SLO output frequency to the local control setting (independent of the local or remote mode setting), and then respond normally. This command does not require a response from the SLO. The command is:

```
RESTART(cr)(lf)
```

30.2.3.3 Fault inputs. There shall be three inputs available for fault processing. Input impedance shall be 1000 ohms. A high level between +2.7 V to +5 V shall represent no fault. A low level between +1 V and 0 V shall represent a fault.

30.2.4 Built-in test (BIT). The synthesizer shall incorporate BIT to the maximum extent practical. As a minimum the BIT shall provide fault status of the high-frequency and low-frequency phase-lock loops, output power level, and the microcontroller.

30.3 Selection of parts, materials, and processes. Unless otherwise specified herein, parts, materials, and processes selected in accordance with MIL-STD-965, MIL-STD-242, and MIL-STD-454 shall be considered as standard and used whenever suitable. The requirements for parts control, the use of military part identifiers, the choice of parts and materials options, and replaceability shall be in accordance with the parts, materials, and processes criteria as specified in MIL-E-16400. Parts, materials, and processes requirements contained herein shall not relieve the contractor of the responsibility of complying with performance and other requirements as specified herein.

30.3.1 Semiconductor devices. Semiconductor devices shall be in accordance with JANTX of MIL-S-19500. Metallic oxide rectifiers shall not be used.

30.3.2 Resistors. Resistors shall be in accordance with MIL-STD-199 and Requirement 33 of MIL-STD-454. Fixed film resistors with resistance values greater than two megaohms shall not be used. Variable resistors shall not be used.

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30.3.3 Capacitors. Capacitors shall be as specified in MIL-STD-198 and Requirement 2 of MIL-STD-454.

30.3.4 Transformers and inductors. Transformers and inductors used in the SLO shall be in accordance with Requirement 14 of MIL-STD-454.

30.3.5 Quartz crystal units. Quartz crystal units shall be in accordance with Requirement 38 of MIL-STD-454.

30.3.6 Electrical filters. Electrical filters used in the SLO shall be as specified in Requirement 70 of MIL-STD-454.

30.3.7 Switches. Switches shall be in accordance with Requirement 58 of MIL-STD-454. The frequency change switches shall be located on the top edge of the module. The switches shall display the operating frequency of the receiver. The receiver operating frequency is 19.95 MHz higher than the operating frequency of the SLO.

30.3.8 Sockets, shields, and clamps. Sockets and accessories for plug-in parts and the use of shielding shall be in accordance with Requirement 60 of MIL-STD-454.

30.3.9 Terminals. Terminals, terminal boards, and strips shall be as specified in Requirement 19 of MIL-STD-454. No more than three wires shall be terminated in an individual terminal or plug unless approval is obtained from the procuring activity. Terminal boards shall be secured only by machine screws and be capable of ready removal.

30.3.10 Wire, hookup. The criteria for the selection and application of equipment internal hookup wire shall be as specified in Requirement 20 of MIL-STD-454. Bare wire shall not be used.

30.3.11 Cable, coaxial RF. Coaxial RF cable selection and use shall be in accordance with Requirement 65 of MIL-STD-454.

30.3.12 Cable, multiconductor. Either solid or stranded multiconductor cable shall be selected and used within the equipment in accordance with Requirement 66 of MIL-STD-454.

30.3.13 Flammable materials. Flammable materials shall not be used in construction for either electrical insulation or mechanical purposes as specified in Requirement 3 of MIL-STD-454.

30.3.14 Insulating materials, electrical. All electrical insulating materials employed to construct the system shall be selected in accordance with Requirement 11 of MIL-STD-454.

30.3.15 Adhesives. If used, adhesives shall be in accordance with Requirement 23 of MIL-STD-454.

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30.3.16 Metals and treatments. Metals shall be of corrosion-resistant types or so processed as specified in the Corrosion-protection/corrosion-resisting treatments paragraph of MIL-E-16400 and in accordance with Requirement 15 and Requirement 16 of MIL-STD-454.

30.3.17 Finish. The finishes used for the SLO enclosure and parts shall be as specified in the Painting paragraph of MIL-E-16400.

30.3.18 Connectors. All electrical connections to the SLO shall be via connectors.

30.3.18.1 Signal connector. The DC power, remote control, and fault status signals shall be input on a 25 contact M24308 connector receptacle that shall be intermateable with the existing local oscillator power connector in the AN/SSR-1A combiner-demodulator. Connector pin assignments shall be as specified in TABLE VII.

TABLE VII. Connector pin assignments.

Pin	Signal
1	+20V
2	GND
3	-20V
4	Not used
5	SLO Fault OUT
6	Not used
7	Fault A IN
8	Fault B IN
9	Fault C IN
10	Not used
11	Serial Data Input A
12	Not used
13	Serial Data Output
14	Not used
15	Not used
16	Not used
17	Not used
18	Not used
19	Not used
20	Not used
21	Not used
22	Not used
23	Serial Data Input B
24	Not used
25	Not used

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30.3.18.2 RF connector. The RF connector shall be an SMA bulkhead mount jack in accordance with the requirements of MIL-C-39012.

30.4 Frequency adjustment. The frequency adjustment shall be accessible through a hole in the top edge of the module.

30.5 Reliability. The MTBF of the module shall be calculated to exceed 100,000 hours. The contractor shall perform reliability calculations based on an approved contractor's mathematical model using MIL-HDBK-217 for component reliability data.

30.6 Environmental requirements. The SLO shall conform to the environmental requirements specified in 3.5.2 through 3.5.2.12 for below-deck equipment.

30.6.1 Magnetic field. The SLO module shall conform to the magnetic field environmental requirements specified in DoD-STD-1399, Section 070, Part I, paragraph 5.

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